INCH-POUND

MIL-M-28787/269A(SH)
AMENDMENT 1
19 November 1990

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

AMPLIFIER, DIFFERENTIAL, ANALOG, KEY CODE FGH

This amendment forms a part of Military Specification MIL-M-28787/269A(EC), dated 30 August 1982, and is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 4

TABLE I: Delete and substitute new table I as printed on pages 3 and 4 of this amendment.

PAGE 6

4.4.2, third line: Delete "A, B, C" and substitute "C, A, and B".

4.5.1, fifth and sixth lines: Delete "B and D through V" and substitute "D through U".

4.5.3, fifth line: Delete "D through V" and substitute "D through U".

4.5.3, seventh line: Delete "C through V" and substitute "D through U".

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TABLE III: Delete and substitute new table III as printed on page 5 of this amendment.

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TABLE VIII: Delete and substitute new table VIII as printed on page 6 of this amendment.

PAGE 25

TABLE X, test condition K2, C to pin column: Delete "39" and substitute "29".

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TABLE XVII, test condition V4, B to pin column: Delete "5" and substitute "2".

Preparing activity:
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(Project 5963-N783)

Agent: NW

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/–3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		–55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1–A4	V_S stress	II	Q	–90	90	--	--	--	--	--	--	--	--	mV
B1–B8	Output short circuit current	III	S	9	40	9	55	9	40	9	55	9	40	mA
C1	Isolation, frame	IV	S											
C2	Isolation, used to unused	IV	X											
	Initial			10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life			1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
D1,D2	Power supply current	V	S	1.0	12.4	1.0	13.0	1.0	12.0	1.0	14.0	1.0	11.0	mA
E1,E2	Power supply filter capacitance	VI	X	1.65	2.75	1.65	2.75	1.65	2.75	1.65	2.75	1.65	2.75	µF
F1–F16	Resistance	VII	X											
	Initial			0.990	1.010	0.985	1.015	0.985	1.015	0.981	1.019	0.984	1.016	kΩ
	End-of-life			0.985	1.015	0.980	1.020	0.980	1.020	0.976	1.024	0.976	1.024	kΩ
F17–F24	Resistance	VII	X											
	Initial			9.90	10.10	9.85	10.15	9.85	10.15	9.81	10.19	9.84	10.16	kΩ
	End-of-life			9.85	10.15	9.80	10.20	9.80	10.20	9.76	10.24	9.76	10.24	kΩ
G1-G4	Input offset voltage check	VIII	X	–3.0	3.0	–3.5	3.5	–3.5	3.5	–4.0	4.0	–4.0	4.0	mV
H1–H4	Input offset current check	VIII	X	–30	30	–43	43	–37	37	–70	70	–45	45	nA
J1-J4	Input bias current check	IX	X	1	110	1	265	1	110	1	265	1	110	nA
K1-K4	AC gain functional check (Gain = 10)	X	X											
	Initial			9.70	10.20	9.60	10.30	9.60	10.30	9.52	10.38	9.52	10.32	V/V
	End-of-life			9.60	10.30	9.50	10.40	9.50	10.40	9.43	10.48	9.43	10.42	V/V

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
K5-K8	AC gain functional check	X	X											
	(Gain = 1)													
	Initial			0.98	1.02	0.97	1.03	0.97	1.03	0.962	1.038	0.962	1.038	V/V
	End-of-life			0.97	1.03	0.96	1.04	0.96	1.04	0.952	1.048	0.952	1.048	V/V
L1-L4	Small signal bandwidth	X	S	0.438	5.000	0.460	5.000	0.403	5.000	0.493	5.000	0.372	5.000	MHz

See footnotes at end of table.

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TABLE II. V_S stress test.

Test condition	Parameter	Connect	Initial limits 25 ± 5°C, ambient		Unit	Test figure
		A to pin	Min	Max		
A1	V_S stress	18	-90	90	mV	5
A2	V_S stress	37	-90	90	mV	5
A3	V_S stress	8	-90	90	mV	5
A4	V_S stress	27	-90	90	mV	5

TABLE III. Output short circuit current test.

Test condition	Parameter	Connect				Initial limits 25 ± 5°C, ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
B1	Output short circuit current	9	Open	18	10	9	40	mA	6
B2	Output short circuit current	29	Open	37	10	9	40	mA	6
B3	Output short circuit current	12	Open	8	10	9	40	mA	6
B4	Output short circuit current	31	Open	27	10	9	40	mA	6
B5	Output short circuit current	Open	9	10	18	9	40	mA	6
B6	Output short circuit current	Open	29	10	37	9	40	mA	6
B7	Output short circuit current	Open	12	10	8	9	40	mA	6
B8	Output short circuit current	Open	31	10	27	9	40	mA	6

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TABLE VIII. Input offset voltage and input offset current check test. 1/ 2/

Test condition	Parameter	Connect			Initial limits 25 ± 5°C, ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	Input offset voltage check.	15	36	18	-300	300	mV	11
G2	Input offset voltage check.	14	13	37	-300	300	mV	11
G3	Input offset voltage check.	6	25	8	-300	300	mV	11
G4	Input offset voltage check.	26	7	27	-300	300	mV	11
H1	Input offset current check.	15	36	18	-30	30	mV	11
H2	Input offset current check.	14	13	37	-30	30	mV	11
H3	Input offset current check.	6	25	8	-30	30	mV	11
H4	Input offset current check.	26	7	27	-30	30	mV	11

1/ $V_{IO} = \frac{V_{OUT}}{\frac{R6 + R4}{R4}} \approx \frac{V_{OUT}}{100}$

2/ $I_{IO} = \frac{\Delta V_{OUT}}{\text{Gain X } R_{Series}}$

H1 = (G1 - M02 reading)/(100 x 10kΩ)
H2 = (G2 - M02 reading)/(100 x 10kΩ)
H3 = (G3 - M02 reading)/(100 x 10kΩ)
H4 = (G4 - M02 reading)/(100 x 10kΩ)

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